

You will be given Stack Overflow posts divided into sentences. Your task is to annotate sentences that indicate positive or negative sentiment towards any API/(s) & its aspect/(s). Each annotation is a tuple that consists of 4 components – the API towards which the sentiment is targeted, the aspect of the API towards which the sentiment is targeted, the sentiment, & the location of the API mention. We are concerned with three sentiment categories – Positive, Negative, and Neutral. Aspects pertain to these two software properties, Performance and Usability.

Guidelines for Annotation:

- You will be given a Stack Overflow post divided into sentences. Although the annotations will be assigned at the sentence level, you should take into account its context within the Stack Overflow post. You probably don't need to read the whole post to do a given annotation.
- A sample Stack Overflow post:

"Here are my thoughts. URL <http://lucene.apache.org/> [Apache-Lucene] is the only way, I think. I don't know any better search lib. Apache Lucene(TM) is a high-performance, full-featured text search engine library written entirely in Java. It is a technology suitable for nearly any application that requires full-text search, especially cross-platform."

A sample annotation, (API-Aspect-Sentiment-Location_of_API), for the above post is shown below:

Id	Sentence	Tuple
1	Here are my thoughts.	
2	URL http://lucene.apache.org/ [Apache-Lucene] is the only way, I think.	Lucene - Usability - Positive - 0
3	I don't know any better search lib.	Lucene - Usability - Positive - -1
4	Apache Lucene(TM) is a high-performance, full-featured text search engine library written entirely in Java.	Lucene - Performance - Positive - 0
5	It is a technology suitable for nearly any application that requires full-text search, especially cross-platform.	Lucene - Usability - Positive - -1 Lucene - Portability - Positive - -1

- Location_of_API should be 0 if the API name is mentioned in the same sentence, or the distance to the nearest mention: -x for

location in previous sentence x distance away, +x for location in following sentence x distance later. Ex: For Sentence 3 above, it should be “API name/-1”.

- In cases of sentences where no tuple is needed, you should leave it blank.
- For sentences with multiple aspects or multiple APIs, you should provide separate tuples as shown in example sentence 5 above. In such cases, the sentiment might be different for each tuple.

API names(s) Annotations: With respect to APIs, annotate specific API name whenever it is mentioned in the text. API references may include libraries, frameworks, packages, modules, classes, tools, or platforms. If only a general category or class of APIs is mentioned (for example, “Java libraries”) instead of a specific name, do not consider that as an API.

Guidelines for Aspect Annotations: An Aspect has to be from the list below. These aspects & their definitions are modified from the ISO/IEC 25010 Quality Attributes. Each aspect further comprises various sub-characteristics as shown below.

- **Performance Efficiency:** Performance of an API relative to the amount of resources used.

- **Time behavior** - API meets requirements of response, processing times and throughput rates when performing its functions.
- **Resource utilization** - API meets requirements of amounts and types of resources used, when performing its functions.
- **Capacity** - The maximum limits of an API parameter meet requirements.

Ex: Anyway as you said, Jackson has a + in performance and that's very important for

me.

- **Usability:** Refers to the level of ease with which developers can interact with the API and use it to accomplish their tasks. The extent to which an API can be utilized by intended users in a particular usage scenario with effectiveness, efficiency, and satisfaction.
 - ***Learnability** - API enables users to achieve specified goals of learning that in turn helps in using that API with effectiveness, efficiency, freedom from risk and satisfaction.*
 - ***Operability** - API has attributes that make it easy to operate and control.*
 - ***User error protection.** API protects users against making errors.*
 - ***User interface aesthetics** - API enables pleasing and satisfying interaction for the user.*
 - ***Accessibility** - API can be used by people with the widest range of characteristics and capabilities.*
 - ***Installability** - API can be successfully installed and/or uninstalled in a specified environment.*
 - ***Documentation** - API has documentation that makes it easier for users to understand and use that API or makes it difficult.*
 - ***Additional Features/Customizability** - How well the API supports diverse user needs by offering a range of customizable options, configuration options and additional features?*
 - ***Effectiveness + Satisfaction** - How effectively does the API enable its users to achieve intended programming goals? The overall contentment users feel when using the API (user satisfaction, user experience/feedback, overall user sentiment).*
- Ex: Started with JGoodies, and found it to be a bit tough to use.*

Detailed Guidelines for Sentiment Labeling: There are two potential sentiment labels: Positive and Negative. The following constitutes the indicators of Positive and Negative sentiment:

“Positive” if one or more of the following applies: The given text indicates one of the following with respect to an API and its aspect

- Getting correct/expected results using an API.

- Enables/helps to achieve something.
- A problem can be or has been solved.
- There is no fault, error, exception or performance degradation.
- An improvement in performance or other aspects.
- A recommendation on or endorses good/best choices.
- Good/useful qualities of tool/features, etc., such as: provides support for certain features, provides advantages over others, availability of certain features.

“Negative” if one or more of the following applies: The given text indicates one of the following with respect to an API and its aspect

- Phrases/words that are bad/unfavorable.
- Disadvantages of certain features.
- Not being able to achieve something using an API (something cannot be done or achieved).
- Unable to figure out or being doubtful of how to do a certain task using an entity.
- An Error or Exception, a Problem/Failure, a Warning, Discontent, Complaint, Disappointment.
- Certain features not working as expected.
- Changing certain features hampers working code/application.
- API being the cause of the problem.

“Neutral” Sentiment: The given text is informational, descriptive, or do not convey a clear positive or negative opinion with respect to an API.